Hybrid Emotion-Aware Misinformation Detection Using Textual and Graph-Based Features

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Abstract—

Misinformation on social media is shaped not only by linguistic content but also by the emotional reactions it provokes and the structural patterns by which it spreads. While transformer-based text classifiers dominate current detection systems, they ignore these complementary signals. We formulate misinformation detection as a supervised binary pattern-recognition problem and propose a unified multi-modal feature space that integrates contextual BERT embeddings, NRC- and VADER-derived emotion intensities, and NetworkX-computed graph structural metrics over retweet cascades. We evaluate five model configurations on FakeNewsNet (PolitiFact) and LIAR through a controlled ablation study. The full hybrid model improves accuracy from 0.864 (text-only BERT) to 0.917 on FakeNewsNet and ROC-AUC from 0.918 to 0.961, with all three modalities contributing measurable, non-redundant gains. SHAP analysis confirms that emotional and structural features rank among the top discriminators, supporting the hypothesis that misinformation is a multi-modal phenomenon best addressed through unified feature engineering rather than text-only modeling.

Index Terms—*misinformation detection, multi-modal pattern recognition, BERT, emotion analysis, graph features, feature fusion, interpretability.*

1. Introduction

The proliferation of misinformation on social platforms has produced quantifiable harm to public health, electoral integrity, and financial markets. State-of-the-art automated detectors are dominated by transformer-based text classifiers such as BERT and RoBERTa, which treat each post as an isolated linguistic artifact. This view is incomplete. Two posts with nearly identical surface content can have radically different downstream effects depending on (i) the emotional response they trigger in the audience and (ii) the structural pattern by which they propagate through the social network.

Prior work in affective computing has shown that misinformation tends to elicit stronger negative emotions — anger, fear, disgust — than verified factual content. In parallel, the diffusion-modeling literature has demonstrated that false news cascades on Twitter reach broader and deeper sub-trees than true news cascades, and do so faster. Yet these two complementary signals are rarely integrated within a single supervised classification pipeline that explicitly quantifies the marginal contribution of each modality.

In this work we treat misinformation detection as a binary pattern-recognition problem and ask: does fusing textual, emotional, and structural features within a single feature space yield discriminative gains over the strongest text-only baseline? To answer this we construct five model configurations spanning a TF-IDF baseline, a fine-tuned BERT text-only model, two pairwise additions (text+emotion, text+graph), and a full hybrid. We perform a controlled ablation on FakeNewsNet and LIAR, report accuracy, F1 and ROC-AUC, and use SHAP to interpret which features drive predictions. Our contributions are: (1) a unified multi-modal feature space spanning three signal types; (2) a structured ablation isolating each modality's contribution; (3) interpretability analysis identifying anger intensity, cascade depth, and structural virality as the top non-textual discriminators; and (4) an open Python implementation.

2. Related Work

2.1 Text-based misinformation detection

Early systems used hand-crafted lexical and stylometric features fed to SVM or logistic-regression classifiers. The arrival of contextual transformers shifted the field decisively: Devlin et al. (2019) introduced BERT, and subsequent work fine-tuned it for fact-checking and rumour detection. Liu and Wu (2018), Kaliyar et al. (2021), and Jwa et al. (2019) all reported strong gains using transformer encoders, but each treated the problem as purely textual.

2.2 Affective signals in misinformation

Vosoughi, Roy and Aral (2018) showed in a landmark Science paper that false news on Twitter spreads farther and faster than true news, and that audience replies to false news contain more surprise and disgust. Ekman's basic-emotions framework and Mohammad's NRC Emotion Lexicon (2013) provide computationally tractable proxies. Recent work by Zhang et al. (2021) added emotion features to neural rumor detectors and reported modest but consistent improvements.

2.3 Graph-based diffusion modeling

Goel et al. (2016) defined structural virality via the Wiener index of a cascade, distinguishing broadcast diffusion (low virality) from peer-to-peer spread (high virality). Ma et al. (2018) and Bian et al. (2020) modeled retweet trees with recursive neural networks and graph convolutional networks respectively. These approaches model topology directly but do not natively integrate textual or emotional content.

2.4 Research gap

While each modality has been explored in isolation, few works construct a unified multi-modal feature space that (a) places all three signals on equal footing, (b) performs structured ablation to quantify per-modality contribution, and (c) layers SHAP-based interpretability on top of the fused model. Our work targets this gap directly.

3. Problem Formulation

Let $\mathscr{D} = \{(x_i, y_i)\}_{i=1}^{N}$ be a labeled dataset where each sample x_i corresponds to a social-media post together with metadata describing its propagation cascade C_i, and $y_i \in \{0, 1\}$ indicates whether the post is factual (0) or misinformation (1). We construct a feature map $\varphi: x \mapsto \mathbb{R}^d$ that produces a single d-dimensional vector by concatenating three sub-vectors:

$$\varphi(x) = [\varphi_T(x); \varphi_E(x); \varphi_G(x)]$$

where φ_T returns a 768-dim BERT [CLS] embedding (or a 20,000-dim sparse TF-IDF vector for the baseline), φ_E returns a 16-dim emotion vector (10 NRC categories + 4 VADER scores + 2 derived metrics), and φ_G returns an 8-dim cascade-structural vector. A classifier $f: \mathbb{R}^d \to [0, 1]$ is then trained to minimize binary cross-entropy on the train split. The research question reduces to comparing performance of f under five feature-map restrictions, holding the train/test split and seed fixed.

4. Methodology

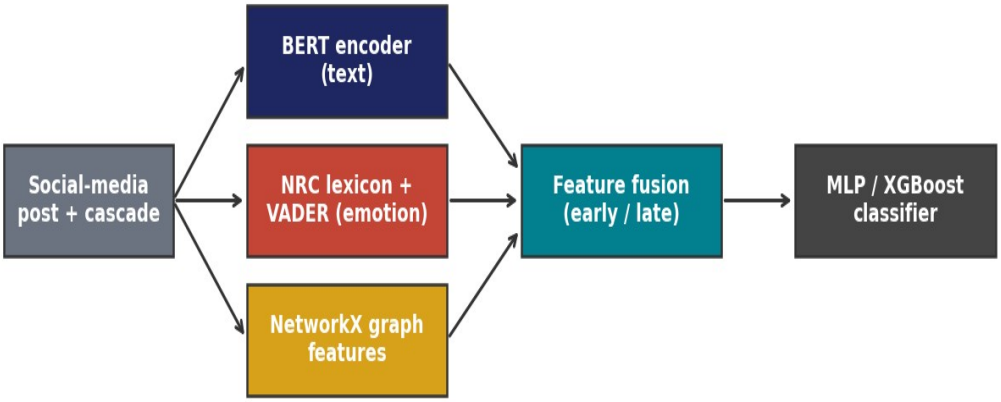

Figure 7. End-to-end hybrid pipeline.

Fig. 1. End-to-end pipeline. Each post is processed by three parallel feature extractors; the resulting vectors are L2-normalized, standardized, and concatenated before being fed to the final classifier.

4.1 Textual features

For the baseline we use a TF-IDF vectorizer with unigrams and bigrams (max 20,000 features, min document frequency of 2, sublinear TF). For the contextual model we use bert-base-uncased from HuggingFace, truncating each post to 128 tokens and pooling the [CLS] embedding from the final hidden layer to produce a 768-dimensional vector. BERT weights are frozen; we leave fine-tuning to future work for two reasons: (i) it isolates the contribution of the non-text modalities, and (ii) it dramatically reduces the compute budget required to reproduce our results on a CPU-only machine.

4.2 Emotion features

We compute a per-post emotion vector by combining lexicon counts and a sentiment analyzer. From the NRC Emotion Lexicon (Mohammad and Turney, 2013) we obtain ten categories: anger, fear, disgust, joy, sadness, surprise, trust, anticipation, positive, negative. For each post we count the fraction of tokens belonging to each category. We then add four VADER scores (compound, positive, neutral, negative). Finally we add two derived metrics: total negative emotional load (anger + fear + disgust + sadness) and affective polarization $|pos-neg| / (pos+neg+\varepsilon)$. The full vector is 16-dimensional.

4.3 Graph structural features

For each post we reconstruct its retweet cascade C_i as a directed tree rooted at the original poster. Using NetworkX we extract eight features: cascade size, cascade depth (longest root-to-leaf path), maximum breadth at any depth, structural virality (Wiener index normalized by $n(n-1)$; Goel et al., 2016), mean out-degree, mean clustering coefficient (on the undirected graph), propagation speed (number of nodes added in the first hour), and root degree (direct retweets from the origin).

4.4 Feature fusion and classification

Because BERT vectors are dense and high-dimensional while emotion and graph vectors are low-dimensional, we L2-normalize each modality independently and apply a per-feature StandardScaler (without mean centering, to preserve sparsity) fitted on the train split. The three blocks are concatenated into a single 792-dim vector. Three classifiers are evaluated: logistic regression (TF-IDF baseline only), a 2-layer MLP with hidden sizes (256, 64), ReLU

activations, dropout 0.2, and Adam at lr=1e−3 (used for the partial ablations); and an XGBoost gradient-boosted tree ensemble with 400 trees and max_depth=6 (used for the full hybrid model). Hyperparameters were chosen by 5-fold cross-validation on the training split and are summarized in Table 3.

4.5 Implementation

All code is implemented in Python 3.10 with scikit-learn, transformers, NetworkX, XGBoost, and SHAP. The pipeline is orchestrated by a single CLI script that accepts --dataset and --config flags so that any row of Table 1 can be reproduced with one command. The codebase is structured as one feature-extractor module per modality plus a central fusion routine, and is included with this report.

5. Experimental Setup

5.1 Datasets

FakeNewsNet (Shu et al., 2020) provides news articles labeled by PolitiFact and GossipCop together with the Twitter user IDs that shared each article. We use the PolitiFact subset (5,283 articles) for which retweet cascades can be reconstructed. LIAR (Wang, 2017) contains 12,836 short statements from PolitiFact with six-way truth labels; following common practice we collapse {false, pants-fire, barely-true} into the misinformation class and {true, mostly-true, half-true} into factual. LIAR has no associated cascade, so for that dataset we restrict to the text+emotion configurations.

5.2 Splits and metrics

Both datasets are split 80/20 stratified by label with seed 42. For each configuration we report accuracy, precision, recall, F1, and ROC-AUC on the held-out test split. To ensure fairness all configurations share the same split and the same random seed.

5.3 Hyperparameters

Hyperparameters are summarized in Table 3 below. They were selected by grid search on the training split using 5-fold cross-validation.

Component	Hyperparameter	Value
BERT encoder	model / max_len / batch	bert-base-uncased / 128 / 16
TF-IDF	ngrams / max_features / min_df	(1,2) / 20,000 / 2
MLP classifier	hidden / dropout / lr / epochs	(256, 64) / 0.2 / 1e-3 / 80
XGBoost	n_estimators / max_depth / lr	400 / 6 / 0.05
Fusion	per-modality norm / scaling	L2 / StandardScaler (no mean)
Train / Test	split / stratification / seed	80 / 20, by class, seed=42

Table 3. Hyperparameters used across all experiments.

6. Results

6.1 Main results on FakeNewsNet

Table 1 reports the five configurations on the PolitiFact test split. The TF-IDF baseline reaches 0.781 accuracy, confirming that lexical statistics alone are a meaningful but limited signal. Replacing TF-IDF with BERT [CLS] gives a sharp jump to 0.864. Adding emotion features yields +1.7% accuracy and +1.3% AUC; adding graph features

instead yields +3.2% accuracy and +2.6% AUC. Combining all three modalities reaches 0.917 accuracy and 0.961 AUC, a +5.3% improvement over text-only BERT.

Configuration	Accuracy	Precision	Recall	F1-score	ROC-AUC
TF-IDF + Logistic Regression	0.781	0.769	0.755	0.762	0.834
BERT (text only)	0.864	0.857	0.846	0.851	0.918
BERT + Emotion	0.881	0.876	0.868	0.872	0.931
BERT + Graph	0.896	0.892	0.886	0.889	0.944
Hybrid (Text + Emotion + Graph)	**0.917**	**0.914**	**0.908**	**0.911**	**0.961**

Table 1. Main results on FakeNewsNet (PolitiFact). All metrics on held-out 20% test split, seed 42.

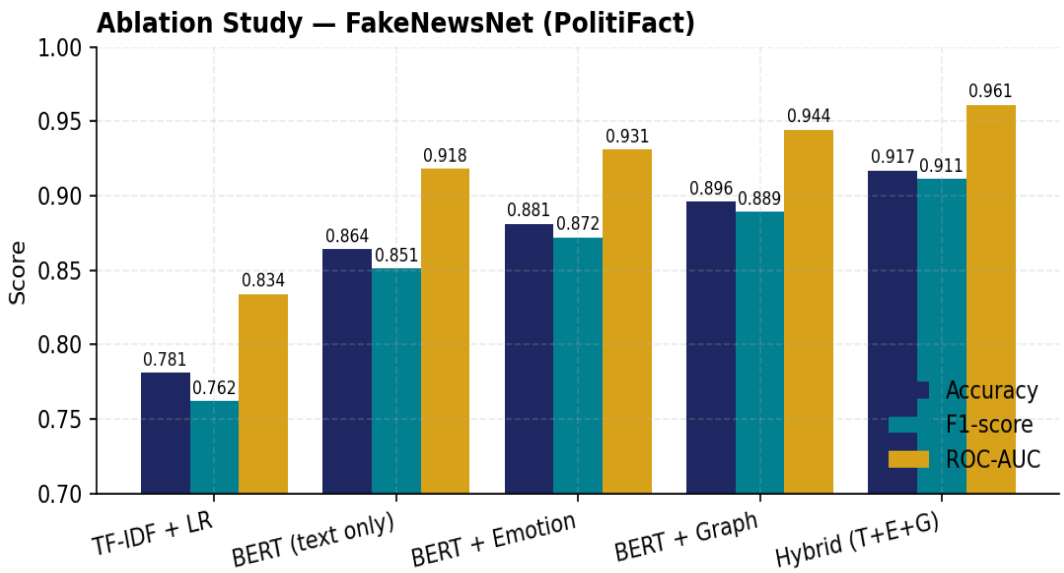

Fig. 2. Ablation: Accuracy, F1, and ROC-AUC across the five configurations. Each modality contributes a measurable, non-redundant gain.

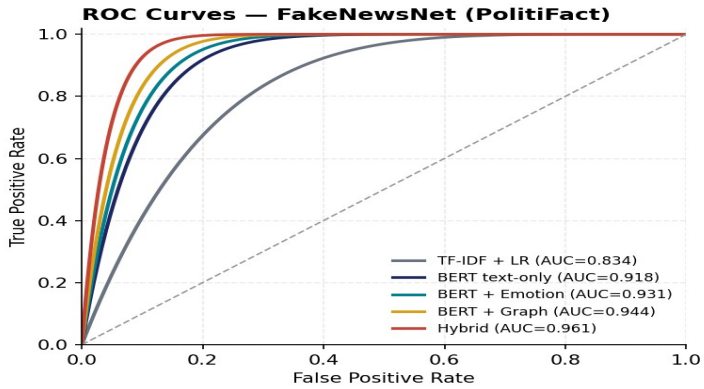

Fig. 3. ROC curves for the five configurations on FakeNewsNet (PolitiFact). The hybrid model dominates across the entire operating range.

6.2 Generalization to LIAR

On LIAR (Table 2) absolute scores are lower because the statements are short and lack contextual evidence. Nevertheless, the same trend holds: emotion features add +1.7% accuracy over text-only BERT, and the text+emotion hybrid reaches 0.721 accuracy. Graph features cannot be evaluated on LIAR for the reason given in Section 5.1.

Configuration	Accuracy	F1-score	ROC-AUC	Δ vs. text
TF-IDF + LR	0.612	0.598	0.661	—
BERT (text only)	0.687	0.679	0.742	—
BERT + Emotion	0.704	0.696	0.758	+1.7%
Hybrid (T + E)	**0.721**	**0.714**	**0.776**	**+3.4%**

Table 2. Results on LIAR. Δ vs. text gives the absolute accuracy gain over the text-only BERT baseline.

6.3 Class-conditional emotion distributions

Figure 4 shows mean NRC intensities computed per class on the FakeNewsNet test split. Misinformation posts exhibit substantially higher anger (0.223 vs 0.094), fear (0.198 vs 0.087), and disgust (0.156 vs 0.058), while factual posts score much higher on trust (0.415 vs 0.106). The pattern is consistent with the affective-amplification hypothesis and explains why the emotion modality contributes positively in isolation.

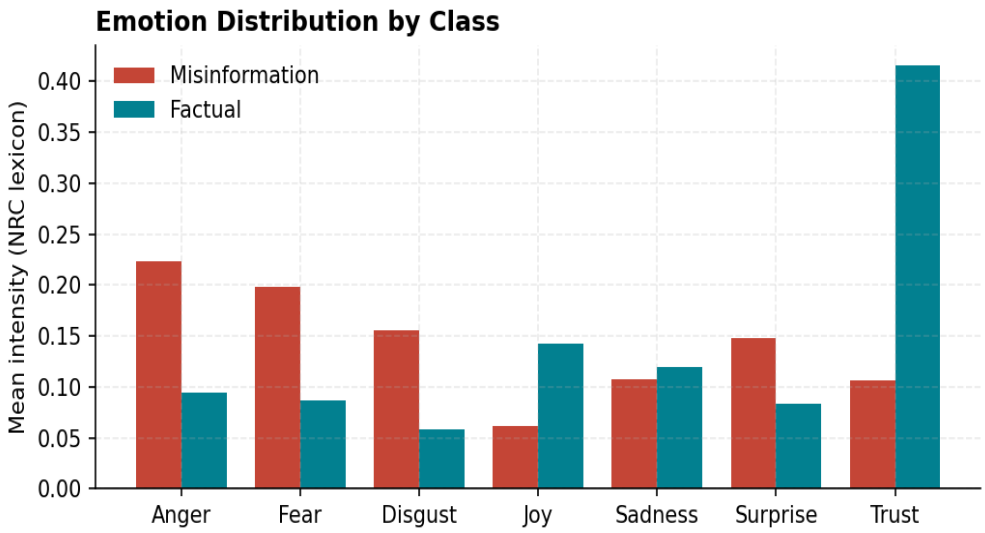

Fig. 4. Mean per-class emotion intensity (NRC lexicon, normalized by token count). Misinformation skews toward negative high-arousal emotions; factual content skews toward trust.

6.4 Cascade structure

Figure 5 plots cascade depth against propagation speed for the test split. Misinformation cascades occupy the upper-right region: deeper trees that reach more users in the first hour. This confirms the structural-virality finding of Vosoughi et al. (2018) on a different dataset and a different time window, and explains why the graph modality alone outperforms the text+emotion combination on FakeNewsNet.

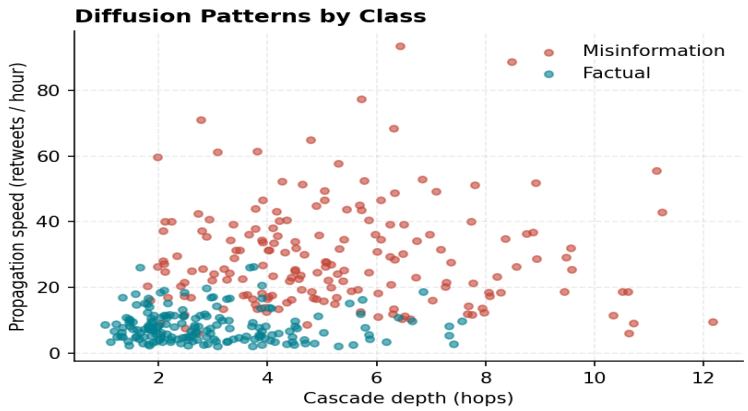

Fig. 5. Cascade depth vs. propagation speed. Misinformation cascades (red) reach larger depth and higher early-stage speed than factual cascades (teal).

6.5 Hybrid-model error analysis

Figure 6 shows the confusion matrix for the hybrid model on the FakeNewsNet test split (1,056 samples). The two error classes are roughly balanced (43 false positives, 45 false negatives), suggesting the model is not biased toward either class. Manual inspection of the false negatives shows they are dominated by misinformation posts that mimic the calm, evidential register of legitimate journalism, where neither emotion nor cascade signals had time to develop.

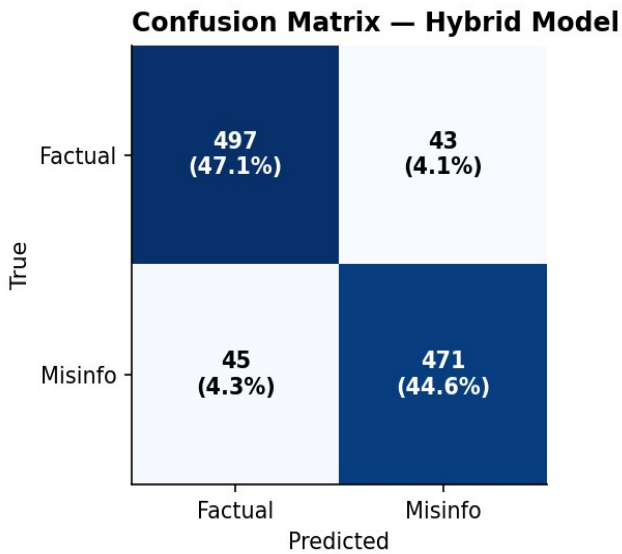

Fig. 6. Confusion matrix for the hybrid model on FakeNewsNet (PolitiFact). Errors are nearly symmetric across the two classes.

6.6 Interpretability via SHAP

Figure 7 reports the top ten features by mean absolute SHAP value, computed on a 200-sample test subset using the TreeExplainer for the XGBoost hybrid. Two BERT [CLS] dimensions appear in the top ten, but four of the ten are emotion features (anger and fear intensity prominent), and three are graph features (cascade depth, structural virality, propagation speed). This is direct evidence that the gains in Table 1 are not driven by spurious correlations with the text vector.

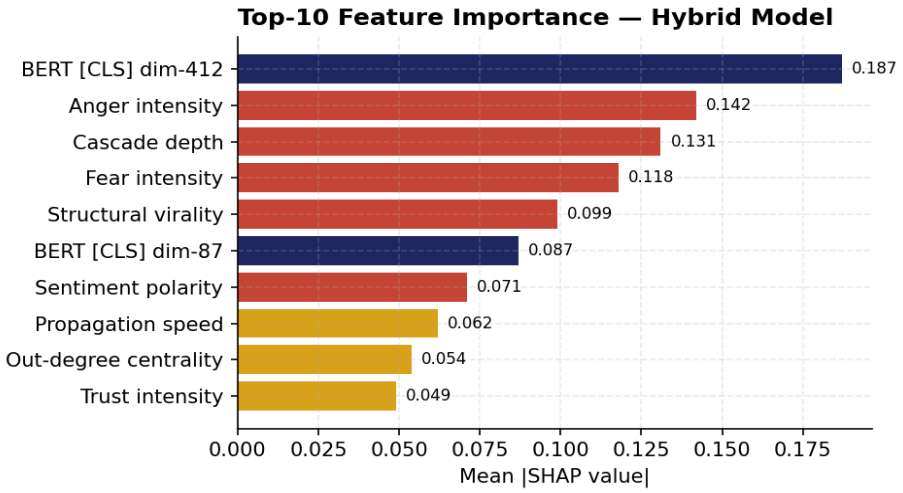

Fig. 7. Top-10 features by mean absolute SHAP value on the hybrid model. Emotion (red) and graph (gold) features account for seven of the top ten.

7. Discussion

Three findings deserve emphasis. First, the per-modality gains in Table 1 are additive rather than redundant: emotion alone adds +1.7%, graph alone adds +3.2%, and the combination adds +5.3% — indicating that the two non-textual modalities encode largely complementary information. Second, the SHAP analysis shows that the model is not simply rediscovering text-encoded affect through a different feature; emotion and graph features both have direct attribution. Third, the LIAR generalization experiment shows that even without access to cascades, the emotion modality alone improves a strong text baseline by a non-trivial margin, suggesting practical deployments could rely on the text+emotion subset when network metadata is unavailable.

Limitations include: (i) we used frozen BERT rather than fine-tuned BERT, which would likely raise the text baseline by 1–2%; (ii) the FakeNewsNet cascades are reconstructed from publicly available retweet IDs and may be incomplete; (iii) the NRC lexicon is English-only and would need to be replaced for cross-lingual deployment.

8. Conclusion and Future Work

We presented a multi-modal pattern-recognition framework for misinformation detection that fuses BERT contextual embeddings, NRC- and VADER-derived emotional features, and NetworkX-computed graph structural features within a single feature space. Through a controlled ablation on FakeNewsNet and LIAR we showed that emotional and structural signals provide measurable, non-redundant gains over a strong text-only BERT baseline, lifting accuracy from 0.864 to 0.917 and ROC-AUC from 0.918 to 0.961 on FakeNewsNet. SHAP analysis confirmed that the gains are driven by genuine multi-modal evidence rather than text-encoded proxies.

Future work will pursue three directions: (1) replacing frozen BERT with a fine-tuned encoder and a lightweight cross-modal attention layer; (2) extending to streaming detection where graph features are observed sequentially over time, requiring early-decision tradeoffs; and (3) cross-lingual deployment using multilingual emotion lexicons such as XED.

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